Technology Stack

Date	1 July 2026
Team ID	SWTID-2026-7714
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	React.js, HTML5, CSS3, JavaScript	Provides a responsive and interactive user interface for applicants and admins.
2	**Backend / Server-Side**	Python (Flask)	Lightweight web framework to build RESTful APIs and handle business logic.
3	**Machine Learning Model**	Scikit-learn	Used to build, train, and evaluate the credit card approval prediction model.
4	**Data Processing & Analysis**	Pandas, NumPy	Efficient data manipulation, cleaning, and feature engineering.
5	**Database**	MySQL	Relational database to store applicant details, application data, and predictions.
6	**API / External Services**	Credit Bureau API (Example: CIBIL API)	Fetch credit score and credit history for applicants.
7	**Deployment / Hosting**	Render / Heroku (Backend) Vercel / Netlify (Frontend)	Cloud platforms for easy deployment, hosting, and scalability.
8	**Version Control**	Git, GitHub	For source code management, collaboration, and tracking changes.